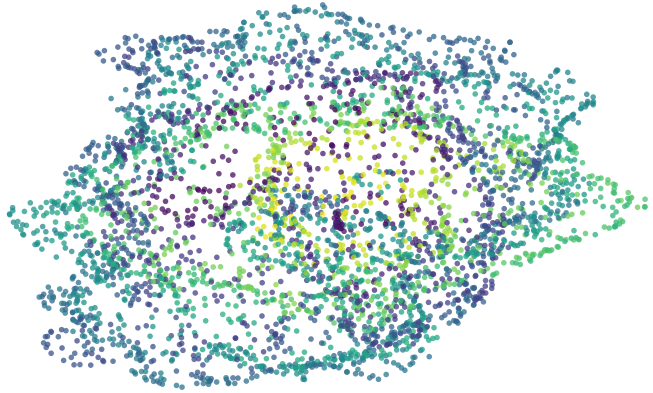
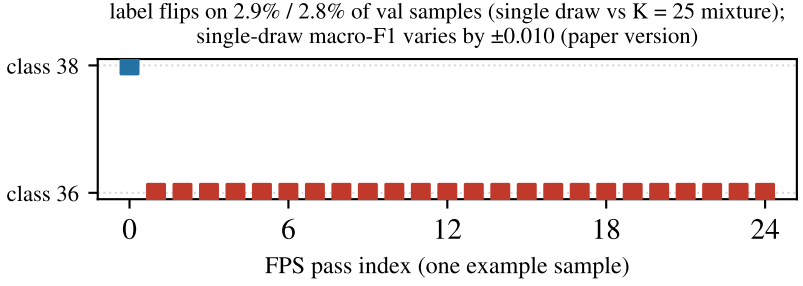
Seed sensitivity of farthest-point sampling: one unseeded draw makes inference nondeterministic — averaging K = 25 passes removes it

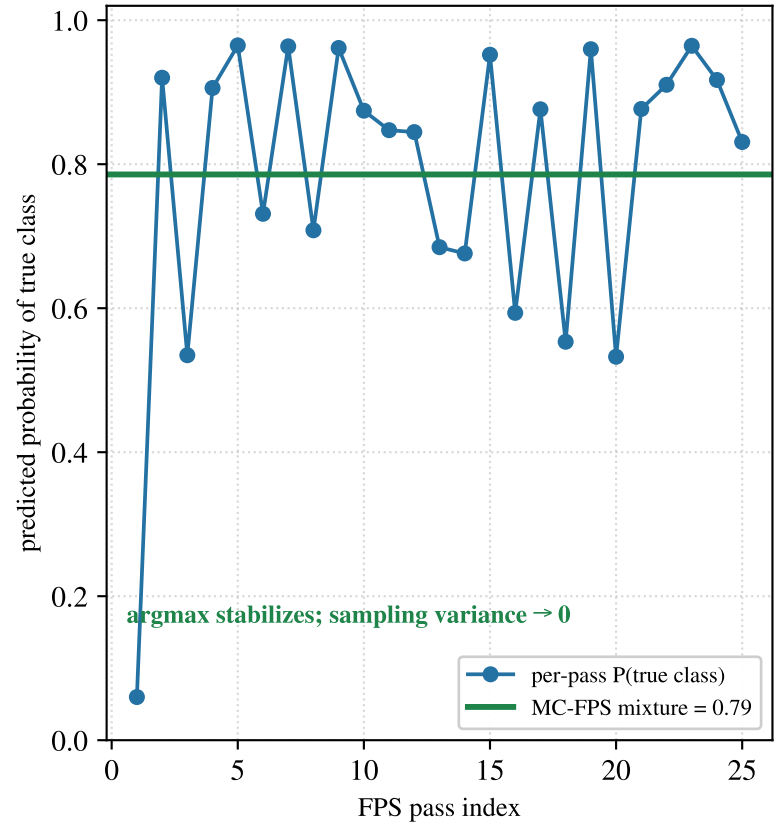
Same input, 25 FPS passes (unseeded initial centroid)



one protein surface point cloud



MC-FPS: average the 25 probability vectors (no retraining, no code change)



Decision criterion and consequences

Fix the seed

you control the code path (bit-exact reproducibility)

Average over seeds

released checkpoints or selective prediction

Consequences

val macro-F1 (25-draw mean → K25): 0.867→0.888 / 0.873→0.890

paired bootstrap vs 25-draw mean: P < .001 / .005 (val)

paper-test macro gain within noise (P = .212) — null

AURC (risk–coverage): −32% / −22% (val)

reproduces on PointNet++ SSG/MSG (ModelNet40; earlier audit)

2 of 2 deep configs reverse vs 3D-SURFER: 7/50 draw reversals